



Review

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Role of late-night eating in circadian disruption and depression: a review of emotional health impacts

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[Purpose] Circadian rhythms regulate key physiological and behavioral processes. Modern lifestyles, characterized by irregular schedules, artificial lighting, and late-night eating disrupt these rhythms, leading to metabolic and emotional health disturbances. Late-night eating misaligns the circadian clock, affecting neurotransmitter function, hormonal rhythms, and inflammatory pathways, thereby increasing the risk of mood disorders.

[Methods] This review synthesizes findings from human and animal studies that examined the effects of late-night eating on circadian misalignment and emotional health. Studies were selected on the basis of their relevance to circadian disruption and mood regulation. This review categorizes the physiological mechanisms involved and evaluates dietary timing interventions, such as time-restricted eating (TRE), earlier meal timing, and meal regularity as potential strategies to mitigate these effects.

[Results] Late-night eating delayed melatonin onset, elevated nocturnal cortisol levels, disrupted serotonin and dopamine rhythms, and increased systemic inflammation, contributing to emotional instability. Aligning mealtimes with the circadian rhythm improves sleep quality, neurotransmitter balance, and stress resilience. TRE and earlier meal timing show promise in mitigating these effects, although further research is needed to optimize the strategies for different chronotypes and metabolic profiles.

[Conclusion] Circadian-aligned meal timing is a practical and noninvasive approach to mitigate the adverse effects of late-night eating on metabolic and emotional health. Future research should refine personalized chrononutrition interventions and assess their long-term effects on mood disorders. Implementing meal-timing strategies based on circadian science could serve as a crucial tool for improving public health and preventing diet-related emotional dysregulation.

[Keywords] late-night eating, circadian rhythm, depression, emotional health, time-restricted eating, sleep quality, metabolism, dietary timing, serotonin, melatonin, mood regulation

INTRODUCTION

The circadian rhythm is a fundamental biological clock that governs a wide range of physiological and behavioral processes, including metabolism, sleep, and emotional regulation¹⁻³. This finely tuned system relies on external cues, such as light exposure, sleep-wake cycles, exercise timing, and feeding patterns to maintain environmental alignment¹⁻⁴. The intricate interplay between external environmental cues, the central circadian clock, and peripheral clocks is summarized in Figure 1, which highlights the effect of late-night eating and circadian misalignment on metabolism, sleep, and emotional health. However, modern lifestyles, characterized by irregular schedules, artificial lighting, and late-night eating, increasingly disrupt these rhythms, posing significant risks to both physical and emotional health⁵.

Late-night eating refers to the intake of food after 9 PM or within 2-3 hours of bedtime, accounting for at least 25% of the total daily caloric intake⁶. Late-night eating is a critical factor in circadian misalignment⁷. Disruptions in feeding schedules are associated with metabolic dysregulation, poor sleep quality, and emotional disturbances including depression⁸⁻¹⁰. Despite growing evidence linking meal timing to these adverse outcomes, the precise mechanisms by which late-night eating contributes to emotional health challenges remain unclear.

This review addresses this gap by integrating late-night eating, circadian rhythms, and emotional health findings. By emphasizing the role of late-night eating in circadian disruption and its subsequent impact on mood regulation, this study highlights the potential of meal-timing interventions as a non-pharmacological strategy for improving emotional well-being. Unlike previous reviews that focused solely on metabolic consequences, this review uniquely integrates the impact of late-night eating on both circadian disruption and emotional health.

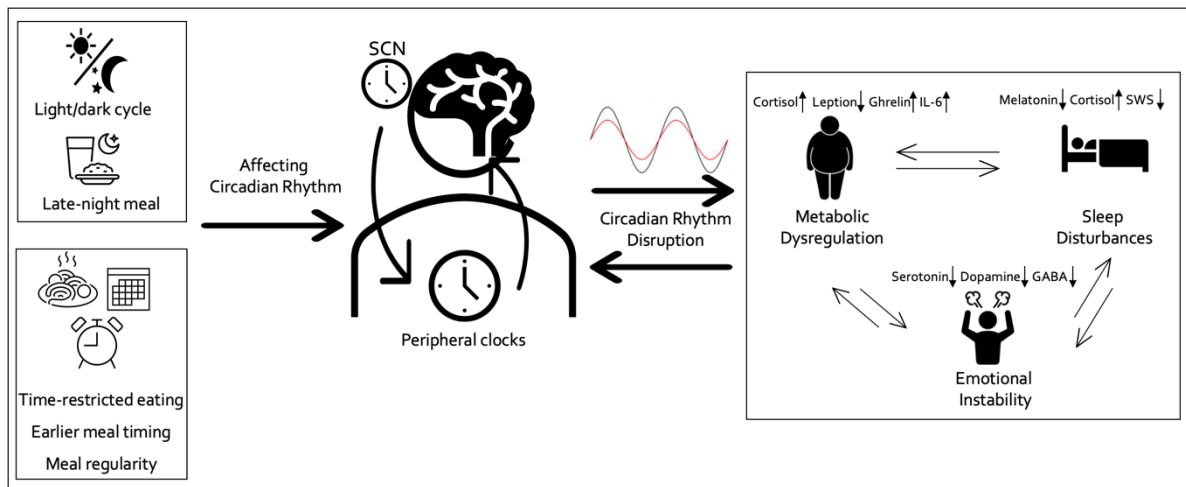


Figure 1. The impact of late-night eating and irregular meal patterns on circadian rhythms and health outcomes. External cues, such as the light/dark cycle and feeding schedules, regulate the central circadian clock (SCN) and peripheral clocks. Disruptions in these rhythms lead to circadian misalignment, which in turn contributes to metabolic dysregulation, sleep disturbances, and emotional instability.

METHODOLOGY

This narrative review examined the relationship between late-night eating, circadian disruption, and emotional health by synthesizing relevant literature from PubMed, Embase, and Web of Science. Rather than relying on a predefined set of keywords, an iterative search strategy was employed to comprehensively identify studies addressing (1) metabolic and circadian effects of late-night eating, (2) effect of circadian misalignment on emotional health, and (3) dietary strategies such as time-restricted eating (TRE) as potential interventions. Given the interdisciplinary nature of this topic, studies were selected based on their conceptual relevance to these interconnected domains, incorporating both human and animal research, where applicable.

CIRCADIAN RHYTHMS, METABOLIC REGULATION, AND MENTAL HEALTH

The circadian rhythm is an intrinsic, approximately 24-hour timekeeping system that regulates a broad spectrum of physiological and behavioral processes, including sleep-wake cycles, hormonal secretion, and metabolism^{11,12}. At its core, this system is governed by the suprachiasmatic nucleus (SCN) of the hypothalamus, often referred to as the “master clock.” The SCN synchronizes with external environmental cues or zeitgebers, such as the light-dark cycle, to ensure that the internal biological rhythms remain aligned with external time¹³.

In addition to light exposure, other modulators such as feeding schedule and sleep pattern play a critical role in circadian alignment. For instance, meal timing has been shown to influence the peripheral clocks located in metabolic organs, such as the liver and pancreas, which operate in con-

cert with the SCN to regulate energy balance and hormonal rhythms¹⁴. Disruptions to these cues, whether from irregular sleep, shift work, or late-night eating, can lead to circadian misalignment with far-reaching health consequences¹⁵.

Circadian misalignment is strongly associated with metabolic dysregulation including insulin resistance, obesity, and type 2 diabetes¹⁶. For example, research has highlighted that individuals with irregular eating schedules are at greater risk of metabolic syndrome and related disorders¹⁷. In addition to metabolic outcomes, circadian rhythms are associated with neurological and emotional health¹⁸. Disruptions in this delicate system can impair the regulation of key neurotransmitters such as serotonin and melatonin, which are critical for mood stabilization¹⁹.

Modern lifestyles, characterized by prolonged artificial lighting, irregular sleep patterns, and an increased prevalence of late-night eating, exacerbate circadian disruptions²⁰. These disruptions not only compromise metabolic and hormonal functions but also contribute to emotional dysregulation. Recognizing the interplay among circadian rhythms, lifestyle behaviors, and health outcomes is essential for developing targeted interventions to mitigate these effects.

DISRUPTION OF CIRCADIAN RHYTHM BY LATE-NIGHT EATING

Meal timing plays a crucial role in the maintenance of circadian alignment. While the SCN primarily responds to light cues, peripheral clocks located in organs such as the liver, pancreas, and gut are particularly sensitive to feeding schedules²¹. These peripheral clocks regulate metabolic processes, including glucose metabolism and lipid storage, and rely on consistent meal timings to synchronize with the central clock. When meal timing becomes irregular—especially

when substantial caloric intake occurs late at night—these peripheral oscillators desynchronize, leading to circadian misalignment^{8,22}.

By delaying melatonin onset and disrupting nighttime metabolic processes, late-night eating further contributes to circadian misalignment, especially when combined with inconsistent sleep patterns²³. Epidemiological studies have indicated that chronic circadian misalignment increases the risk of type 2 diabetes, cardiovascular diseases, and obesity²⁴. Late-night eating exacerbates these risks and impairs the body's ability to maintain homeostasis by disrupting the synchronization between the central and peripheral clocks.

In addition to the metabolic consequences, late-night eating disrupts hormonal rhythms. Irregular feeding patterns reduce nighttime leptin levels and elevate cortisol levels, fostering a physiological state conducive to weight gain and heightened stress²⁵. These disruptions compromise metabolic regulation and increase susceptibility to broader health complications.

As late-night eating has become increasingly prevalent in modern societies, understanding its role in circadian misalignment is crucial. Aligning feeding schedules with the body's natural rhythms has emerged as a critical strategy to safeguard both metabolic and emotional health. This topic is further explored in the context of mood disorders in the following section.

LATE-NIGHT EATING AND ITS IMPACT ON EMOTIONAL HEALTH

Circadian rhythms are intricately linked to emotional health, with disruptions often manifesting as mood disorders including depression and anxiety²⁶. As a driver of circadian misalignment, late-night eating exacerbates these risks by impairing the regulation of mood-related neurotransmitters, hormones, and inflammatory responses.

One of the primary pathways through which late-night eating influences emotional health is the dysregulation of neurotransmitters. Serotonin, a key regulator of mood and emotional stability, is synthesized in alignment with circadian rhythms²⁷. Late-night eating has been associated with delayed serotonin production, which potentially diminishes serotonin availability during critical mood regulation periods. Similarly, melatonin, a hormone essential for sleep and circadian alignment, is suppressed by nocturnal food intake²⁷. This suppression disrupts the sleep quality and creates a feedback loop that exacerbates emotional instability.

Furthermore, late-night eating has been linked to increased levels of systemic inflammation, which is another significant contributor to mood disorders. Studies indicate that irregular meal timing elevates inflammatory markers such as C-reactive protein (CRP) and interleukin-6 (IL-6), both of which are strongly associated with depressive symptoms^{28,29}. These inflammatory responses, combined with metabolic dysregulation, create a physiological environment that increases the susceptibility to mood disorders.

Sleep disturbance, a common outcome of late-night eat-

ing, also plays a critical role in the development of emotional instability. Consuming meals late at night delays the onset of deep sleep stage and reduces overall sleep quality. Poor sleep impairs emotional regulation, contributing to heightened sensitivity to stress and an increased likelihood of depressive episodes³⁰. For example, Kalmbach et al. (2018) found that individuals with circadian misalignment were significantly more likely to report symptoms of depression, with sleep disturbances serving as a mediating factor³¹.

Modern lifestyles increase the risks associated with late-night eating. The prevalence of artificial lighting, irregular schedules, and psychological stress has made late-night eating a widespread habit, further amplifying its adverse effects on mental health³². Individuals who engage in late-night eating within these contexts are particularly vulnerable to circadian disruptions, as their biological rhythms are persistently strained.

Addressing the emotional consequences of late-night eating requires a comprehensive understanding of its physiological and psychological impacts. Targeting dietary timing as a modifiable behavior, interventions such as TRE and synchronized meal schedules are promising non-pharmacological strategies for improving emotional well-being. These approaches not only restore circadian alignment, but also reduce key risk factors for depression, offering critical insights for contemporary mental health research.

MECHANISM LINKING CIRCADIAN DISRUPTION AND EMOTIONAL DISORDERS

Circadian rhythms are strongly associated with the regulation of mood and emotional stability, largely because of their influence on key physiological systems³³. Disruptions to these rhythms, such as those caused by late-night eating, result in a cascade of effects that alter neurotransmitter activity, stress hormone regulation, and systemic inflammation, all of which are critical pathways linking circadian misalignment to emotional disorders³⁴.

Neurotransmitter Dysregulation

The circadian system governs the synthesis and release of neurotransmitters such as serotonin, dopamine, and gamma-aminobutyric acid (GABA), which are essential for mood regulation³⁵. Circadian misalignment, often induced by late-night eating, disrupts the rhythmicity and bioavailability of these neurotransmitters. Serotonin (5-HT) synthesis follows a circadian pattern, peaking during daylight hours, when natural light exposure enhances tryptophan hydroxylase activity³⁶. Late-night eating interferes with this process by altering peripheral metabolic signals, thereby reducing serotonin bioavailability during critical periods of emotional regulation²⁷. Similarly, dopamine release in the striatum and prefrontal cortex follows a diurnal pattern, with peaks in the active phase to support motivation and cognitive function^{37,38}. Late-night eating has been linked to dysregulated dopamine rhythms, potentially increasing susceptibility to anhedonia and depressive symptoms³⁹.

Additionally, GABAergic signaling, which modulates stress responses, is influenced by circadian misalignment and contributes to heightened emotional reactivity⁴⁰.

Hormonal Dysregulation

Stress hormones, particularly cortisol, follow a circadian rhythm that peaks in the morning and declines throughout the day. Late-night eating disrupts this cycle by prolonging cortisol elevation at night, leading to dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis⁴¹. This prolonged stress response impairs emotional resilience and increases susceptibility to anxiety and depression⁴². Melatonin, which is essential for sleep onset and circadian synchronization, is suppressed by nocturnal food intake, delaying the transition to restorative sleep stages and contributing to emotional instability⁴³. Similarly, leptin, an anorexigenic hormone that promotes satiety during sleep, follows a nocturnal peak, but is blunted by late-night eating, whereas ghrelin secretion increases, encouraging continued food intake and metabolic instability⁴⁴. These disruptions reinforce the hyperactivity of the HPA axis, further exacerbating stress responses and mood dysregulation⁴⁵.

Inflammatory Pathways

Chronic circadian misalignment elevates pro-inflammatory cytokines, such as interleukin-6 (IL-6) and tumor necrosis factor-alpha (TNF- α), contributing to neuroinflammation, reduced synaptic plasticity, and impaired neurogenesis^{46,47}. Late-night eating exacerbates this inflammatory response by disrupting the gut microbiota composition, increasing intestinal permeability, and promoting endotoxemia, which further amplifies systemic inflammation⁷. Elevated inflammatory signaling interferes with serotonin metabolism by activating the kynurenine pathway, diverting tryptophan metabolism away from serotonin synthesis toward neurotoxic metabolites, such as quinolinic acid, a mechanism implicated in mood disorders, including major depressive disorder and anxiety⁴⁸⁻⁵⁰. Additionally, inflammation-induced glial activation disrupts neural connectivity, impairing the prefrontal-limbic circuit function that is essential for mood regulation^{50,51}. These findings underscore late-night eating as a modifiable risk factor for inflammation-driven emotional dysregulation. Studies indicate that individuals with heightened inflammation are significantly more likely to experience mood disorders, reinforcing the biological link between circadian disruption and emotional instability⁷.

Sleep and Emotional Health

Sleep disturbances caused by circadian misalignments increase the risk of emotional disorders¹⁷. Poor sleep impairs the brain's ability to regulate emotions, thereby affecting decision making and stress management⁵². This bidirectional relationship between circadian disruption and emotional dysregulation exacerbates the negative impact of late-night eating on mental health⁵³. Neuroimaging studies highlight that this disruption is particularly pronounced in the prefrontal cortex and amygdala, regions essential for emotional regulation^{54,55}. Sleep deprivation has been shown to heighten

amygdala reactivity while weakening prefrontal control, leading to increased emotional volatility and heightened stress sensitivity⁵⁶.

Late-night eating further contributes to sleep disturbances by delaying sleep onset and reducing slow-wave sleep (SWS), a critical stage for emotional processing and memory consolidation^{57,58}. As SWS plays a key role in regulating the HPA axis, its disruption leads to prolonged cortisol elevation, further intensifying stress responses and mood instability^{59,60}.

DIETARY TIMING STRATEGIES FOR CIRCADIAN RHYTHM AND EMOTIONAL REGULATION

Addressing the emotional and physiological consequences of circadian misalignment requires targeted interventions, with dietary timing emerging as a promising non-pharmacological strategy. By realigning feeding schedules with the body's intrinsic rhythms, individuals can mitigate the adverse effects of late-night eating on their emotional well-being and overall health. Key approaches include TRE, earlier meal timing, and meal synchronization with circadian phases.

TRE

TRE, which involves consuming all daily calories within a specific time window (e.g., 8–10 h), has demonstrated significant potential for restoring circadian alignment⁶¹⁻⁶⁵. By confining feeding to daylight hours, TRE reinforces synchronization between the central clock in the SCN and peripheral clocks in metabolic organs⁶⁵⁻⁶⁷. Studies have indicated that TRE improves glucose metabolism and lipid profiles, and enhances sleep quality and mood stability by reducing the misalignment between feeding times and the circadian system. TRE is also associated with reduced nocturnal cortisol levels and stress-related metabolic dysfunction⁶⁸. TRE optimizes glucose utilization and reduces the risk of metabolic disorders by aligning food intake with the circadian peak of insulin sensitivity⁶⁶.

Earlier Meal Timing

Shifting the primary caloric intake to earlier in the day aligns feeding schedule with the body's natural insulin sensitivity, which peaks in the morning and early afternoon⁶⁹. Epidemiological studies suggest that individuals who consume their main meal earlier in the day experience better metabolic outcomes and report lower rates of depressive symptoms^{70,71}. For instance, Cleator et al. (2012) found that late-night eaters had a higher risk of mood disorders than early night eaters, underscoring the psychological benefits of aligning meal timing with circadian biology⁷². Early meal timing facilitates melatonin secretion and improves sleep quality and onset⁶⁹. Despite these benefits, some populations, such as adolescents and shift workers, may require flexible adjustments owing to social and occupational constraints. Gradual meal timing shifts rather than abrupt

changes support adherence and long-term sustainability.

Meal Regularity

Maintaining a stable and predictable meal schedule is essential for reinforcing circadian alignment and mood stability⁷³. Irregular eating patterns, including skipping meals or varying mealtimes, disrupt the peripheral clocks and exacerbate mood instability⁷⁴⁻⁷⁶. Stable mealtimes reinforce the rhythmicity of serotonin and dopamine, which are essential for mood stability⁷⁷. Structured meal planning and self-monitoring help maintain consistent meal patterns, which should be emphasized in dietary guidelines along with nutritional quality.

Integrating Chrononutrition Principles

Chrononutrition focuses not only on maintaining regular meal timing, but also on aligning food intake with circadian metabolic rhythms to optimize energy balance and physiological functions⁷⁸. Avoiding late-night meals, particularly those high in fat and sugar, may reduce metabolic burden and enhance sleep quality, indirectly benefiting emotional stability. Structured meal timing that aligns with circadian metabolic rhythms can support energy regulation and optimize physiological functions⁷⁹.

CONCLUSION

Circadian rhythms are fundamental for maintaining both physiological and emotional homeostasis; however, modern lifestyles characterized by irregular schedules, prolonged artificial lighting, and late-night eating increasingly disrupt these rhythms. This review highlights how late-night eating contributes to circadian misalignment, impairing metabolic and hormonal functions while exacerbating mood dysregulation, particularly depression. By elucidating the mechanisms that link dietary timing to mood regulation, we underscore meal timing as a modifiable lifestyle factor with significant implications for both physical and mental health. Interventions such as TRE, earlier meal scheduling, and consistent meal patterns represent promising and noninvasive strategies to realign circadian rhythms and reduce the risk of mood disorders.

Despite these findings, important knowledge gaps remain in the literature. Although many studies have examined the effects of late-night eating on circadian rhythms and emotional health independently, few have experimentally validated their combined effects in a comprehensive manner. Additionally, although preliminary evidence suggests the potential benefits of TRE and earlier meal timing for stress regulation and sleep quality, further well-controlled trials are required to confirm their efficacy in mitigating mood disorders.

Thus, future research should investigate how dietary timing interacts with sleep architecture, neurotransmitter balance, and inflammatory processes over the long term. Practical solutions are needed to improve adherence to circadian-aligned meal timings in real-world settings. One

promising approach is the integration of digital health technologies such as wearable devices and mobile applications, which may enhance adherence by offering real-time feedback and personalized recommendations.

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REFERENCES

- Lewis P, Oster H, Korf HW, Foster RG, Erren TC. Food as a circadian time cue-evidence from human studies. *Nat Rev Endocrinol*. 2020;16:213-23.
- Bae SA, Fang MZ, Rustgi V, Zarbl H, Androulakis IP. At the interface of lifestyle, behavior, and circadian rhythms: metabolic implications. *Front Nutr*. 2019;6:132.
- Koronowski KB, Sassone-Corsi P. Communicating clocks shape circadian homeostasis. *Science*. 2021;371:0951.
- Lee K, Hong KS, Park J, Park W. Readjustment of circadian clocks by exercise intervention is a potential therapeutic target for sleep disorders: a narrative review. *Phys Act Nutr*. 2024;28:35-42.
- Pot GK. Sleep and dietary habits in the urban environment: the role of chrono-nutrition. *Proc Nutr Soc*. 2018; 77:189-98.
- Jaffar HM, Ahsan W, Aqeel K, Sajjad R, Tariq M, Ahmed H, Fatima N. Effect of late-night eating on health: a survey from university students. *Agric Sci J*. 2023;5:93-9.
- Ni Y, Wu L, Jiang J, Yang T, Wang Z, Ma L, Zheng L, Yang X, Wu Z, Fu Z. Late-night eating-induced physiological dysregulation and circadian misalignment are accompanied by microbial dysbiosis. *Mol Nutr Food Res*. 2019;63:1900867.
- Pickel L, Sung HK. Feeding rhythms and the circadian regulation of metabolism. *Front Nutr*. 2020;7:39.
- Potter GD, Skene DJ, Arendt J, Cade JE, Grant PJ, Hardie LJ. Circadian rhythm and sleep disruption: causes, metabolic consequences, and countermeasures. *Endocr Rev*. 2016;37:584-608.
- Salgado-Delgado R, Tapia Osorio A, Saderi N, Escobar C. Disruption of circadian rhythms: a crucial factor in the etiology of depression. *Depress Res Treat*. 2011;2011:839743.
- Panda S. Circadian physiology of metabolism. *Science*. 2016;354:1008-15.
- Turek FW. Circadian rhythms. *Recent Prog Horm Res*. 1994;49:43-90.
- Pevet P, Challet E. Melatonin: both master clock output and internal time-giver in the circadian clocks network. *J Physiol Paris*. 2011;105:170-82.
- Engin A. Misalignment of circadian rhythms in diet-induced obesity. *Adv Exp Med Biol*. 2024;1460:27-71.
- Nohara K, Yoo SH, Chen ZJ. Manipulating the circadian and sleep cycles to protect against metabolic disease. *Front Endocrinol*. 2015;6:35.
- Depner CM, Stothard ER, Wright KP. Metabolic consequences of

- sleep and circadian disorders. *Curr Diab Rep*. 2014;14:507.
17. Baron KG, Reid KJ. Circadian misalignment and health. *Int Rev Psychiatry*. 2014;26:139-54.
 18. Foster RG. Sleep, circadian rhythms and health. *Interface Focus*. 2020;10:20190098.
 19. Daut RA, Fonken LK. Circadian regulation of depression: a role for serotonin. *Front Neuroendocrinol*. 2019;54:100746.
 20. Ng CM, Kaur S. Environmental light exposure and mealtime regularity: implications for human health. *Chronobiol Int*. 2022;39:1183-94.
 21. Engin A. Circadian rhythms in diet-induced obesity. *Adv Exp Med Biol*. 2017;960:19-52.
 22. Ruddick-Collins LC, Morgan PJ, Johnstone AM. Mealtime: a circadian disruptor and determinant of energy balance? *J Neuroendocrinol*. 2020;32:e12886.
 23. Allison KC, Goel N, Ahima RS. Delayed timing of eating: impact on weight and metabolism. *Curr Obes Rep*. 2014;3:91-100.
 24. Palomar-Cros A, Andreeva VA, Fezeu LK, Julia C, Bellicha A, Kesse-Guyot E, Hercberg S, Romaguera D, Kogevinas M, Touvier M, Srouf B. Dietary circadian rhythms and cardiovascular disease risk in the prospective NutriNet-Santé cohort. *Nat Commun*. 2023;14:7899.
 25. Spiegel K, Leproult R, L'Hermite-Balériaux M, Copinschi G, Penev PD, Van Cauter E. Leptin levels are dependent on sleep duration: relationships with sympathovagal balance, carbohydrate regulation, cortisol, and thyrotropin. *J Clin Endocrinol Metab*. 2004;89:5762-71.
 26. Walker WH 2nd, Walton JC, DeVries AC, Nelson RJ. Circadian rhythm disruption and mental health. *Transl Psychiatry*. 2020;10:28.
 27. Versteeg RI, Serlie MJ, Kalsbeek A, la Fleur SE. Serotonin, a possible intermediate between disturbed circadian rhythms and metabolic disease. *Neuroscience*. 2015;301:155-67.
 28. Milanesechi Y, Kappelman N, Ye Z, Lamers F, Moser S, Jones PB, Burgess S, Penninx BWJH, Khandaker GM. Association of inflammation with depression and anxiety: evidence for symptom-specificity and potential causality from UK Biobank and NESDA cohorts. *Mol Psychiatry*. 2021;26:7393-402.
 29. Nishutry NL, Khandaker MMH, Karmoker JR, Ferdous S, Shahriar M, Qusar M, Islam MS, Kadir MF, Islam MR. Evaluation of serum interleukin-6 and C-reactive protein levels in drug-naïve major depressive disorder patients. *Cureus*. 2019;11:e3868.
 30. Wang W, Zhu Y, Yu H, Wu C, Li T, Ji C, Jiang Y, Ding D. The impact of sleep quality on emotion regulation difficulties in adolescents: a chained mediation model involving daytime dysfunction, social exclusion, and self-control. *BMC Public Health*. 2024;24:1862.
 31. Kalmbach DA, Anderson JR, Drake CL. The impact of stress on sleep: pathogenic sleep reactivity as a vulnerability to insomnia and circadian disorders. *J Sleep Res*. 2018;27:e12710.
 32. Zou H, Zhou H, Yan R, Yao Z, Lu Q. Chronotype, circadian rhythm, and psychiatric disorders: recent evidence and potential mechanisms. *Front Neurosci*. 2022;16:811771.
 33. Cisse YM, Borniger JC, Nelson RJ. Hormones, circadian rhythms, and mental health. *Oxford University Press*. 2019.
 34. Meléndez-Fernández OH, Liu JA, Nelson RJ. Circadian rhythms disrupted by light at night and mistimed food intake alter hormonal rhythms and metabolism. *Int J Mol Sci*. 2023;24:3392.
 35. Kiehn JT, Faltraco F, Palm D, Thome J, Oster H. Circadian clocks in the regulation of neurotransmitter systems. *Pharmacopsychiatry*. 2023;56:108-17.
 36. Cuesta M, Mendoza J, Clesse D, Pévet P, Challet E. Serotonergic activation potentiates light resetting of the main circadian clock and alters clock gene expression in a diurnal rodent. *Exp Neurol*. 2008;210:501-13.
 37. Kim J, Jang S, Choe HK, Chung S, Son GH, Kim K. Implications of circadian rhythm in dopamine and mood regulation. *Mol Cells*. 2017;40:450-6.
 38. Radwan B, Liu H, Chaudhury D. The role of dopamine in mood disorders and the associated changes in circadian rhythms and sleep-wake cycle. *Brain Res*. 2019;1713:42-51.
 39. Grippo RM, Tang Q, Zhang Q, Chadwick SR, Gao Y, Altherr EB, Sipe L, Purohit AM, Purohit NM, Sunkara MD, Cios KJ, Sidikpramana M, Spano AJ, Campbell JN, Steele AD, Hirsh J, Deppmann CD, Wu M, Scott MM, Guler AD. Dopamine signaling in the suprachiasmatic nucleus enables weight gain associated with hedonic feeding. *Curr Biol*. 2020;30:1352-5.
 40. Ono D, Honma Ki, Honma S. GABAergic mechanisms in the suprachiasmatic nucleus that influence circadian rhythm. *J Neurochem*. 2021;157:31-41.
 41. Van Drunen R, Eckel-Mahan K. Circadian rhythms of the hypothalamus: from function to physiology. *Clocks Sleep*. 2021;3:189-226.
 42. Nicolaides NC, Charmandari E, Chrousos GP, Kino T. Circadian endocrine rhythms: the hypothalamic-pituitary-adrenal axis and its actions. *Ann N Y Acad Sci*. 2014;1318:71-80.
 43. McHill AW, Phillips AJK, Czeisler CA, Keating L, Yee K, Barger LK, Garaulet M, Scheer FAJL, Klerman EB. Later circadian timing of food intake is associated with increased body fat. *Am J Clin Nutr*. 2017;106:1213-9.
 44. Taheri S, Lin L, Austin D, Young T, Mignot E. Short sleep duration is associated with reduced leptin, elevated ghrelin, and increased body mass index. *PLoS Med*. 2004;1:e62.
 45. Birketvedt GS, Sundsfjord J, Florholmen JR. Hypothalamic-pituitary-adrenal axis in the night eating syndrome. *Am J Physiol Endocrinol Metab*. 2002;282:E366-9.
 46. Irwin MR, Opp MR. Sleep health: reciprocal regulation of sleep and innate immunity. *Neuropsychopharmacology*. 2017;42:129-55.
 47. Dantzer R, O'Connor JC, Freund GG, Johnson RW, Kelley KW. From inflammation to sickness and depression: when the immune system subjugates the brain. *Nat Rev Neurosci*. 2008;9:46-56.
 48. Bhat A, Pires AS, Tan V, Babu Chidambaram S, Guillemin GJ. Effects of sleep deprivation on the tryptophan metabolism. *Int J Tryptophan Res*. 2020;13:1178646920970902.
 49. Haq S, Grondin JA, Khan WI. Tryptophan-derived serotonin-ky-nurenine balance in immune activation and intestinal inflammation. *FASEB J*. 2021;35:e21888.
 50. Haroon E, Miller AH, Sanacora G. Inflammation, glutamate, and glia: a trio of trouble in mood disorders. *Neuropsychopharmacology*. 2017;42:193-215.
 51. Guo B, Zhang M, Hao W, Wang Y, Zhang T, Liu C. Neuroinflammation mechanisms of neuromodulation therapies for anxiety and depression. *Transl Psychiatry*. 2023;13:5.
 52. Killgore WDS. Effects of sleep deprivation on cognition. *Prog Brain Res*. 2010. p.105-29.
 53. Meyer N, Lok R, Schmidt C, Kyle SD, McClung CA, Cajochen C, Scheer FAJL, Jones MW, Chellappa SL. The sleep-circadian in-

- terface: a window into mental disorders. *Proc Natl Acad Sci U S A*. 2024;121:e2214756121.
54. Pereira SIR, Tsimpanouli ME, Hutchison I, Schneider J, Anderson IM, McFarquhar M, Elliott R, Lewis PA. Cueing emotional memories during slow wave sleep modulates next-day activity in the orbitofrontal cortex and the amygdala. *Neuroimage*. 2022;253:119120.
 55. Chuah LYM, Dolcos F, Chen AK, Zheng H, Parimal S, Chee MWL. Sleep deprivation and interference by emotional distracters. *Sleep*. 2010;33:1305-13.
 56. Killgore WDS. Self-reported sleep correlates with prefrontal-amygdala functional connectivity and emotional functioning. *Sleep*. 2013;36:1597-608.
 57. Lopes TdVC, Borba ME, Lopes RdVC, Fisberg RM, Paim SL, Teodoro VV, Zimberg IZ, Crispim CA. Eating late negatively affects sleep pattern and apnea severity in individuals with sleep apnea. *J Clin Sleep Med*. 2019;15:383-92.
 58. Walker MP. The role of slow wave sleep in memory processing. *J Clin Sleep Med*. 2009;5:S20-6.
 59. Léger D, Debellemanni E, Rabat A, Bayon V, Benchenane K, Chennaoui M. Slow-wave sleep: from the cell to the clinic. *Sleep Med Rev*. 2018;41:113-32.
 60. Goldschmied JR, Cheng P, Hoffmann R, Boland EM, Deldin PJ, Armitage R. Effects of slow-wave activity on mood disturbance in major depressive disorder. *Psychol Med*. 2019;49:639-45.
 61. Manoogian ENC, Chow LS, Taub PR, Laferrière B, Panda S. Time-restricted eating for the prevention and management of metabolic diseases. *Endocr Rev*. 2022;43:405-36.
 62. Queiroz JdN, Macedo RCO, Tinsley GM, Reischak-Oliveira A. Time-restricted eating and circadian rhythms: the biological clock is ticking. *Crit Rev Food Sci Nutr*. 2021;61:2863-75.
 63. Wilkinson MJ, Manoogian EN, Zadourian A, Lo H, Fakhouri S, Shoghi A, Wang X, Fleischer JG, Navlakha S, Panda S, Taub PR. Ten-hour time-restricted eating reduces weight, blood pressure, and atherogenic lipids in patients with metabolic syndrome. *Cell Metab*. 2020;31:92-104.
 64. Regmi P, Heilbronn LK. Time-restricted eating: benefits, mechanisms, and challenges in translation. *iScience*. 2020;23:101161.
 65. Zarrinpar A, Chaix A, Panda S. Daily eating patterns and their impact on health and disease. *Trends Endocrinol Metab*. 2016;27:69-83.
 66. Chaix A, Manoogian EN, Melkani GC, Panda S. Time-restricted eating to prevent and manage chronic metabolic diseases. *Annu Rev Nutr*. 2019;39:291-315.
 67. Oike H, Oishi K, Kobori M. Nutrients, clock genes, and chrononutrition. *Curr Nutr Rep*. 2014;3:204-12.
 68. Jamshed H, Beyl RA, Della Manna DL, Yang ES, Ravussin E, Peterson CM. Early time-restricted feeding improves 24-hour glucose levels and affects markers of the circadian clock, aging, and autophagy in humans. *Nutrients*. 2019;11:1234.
 69. BaHammam AS, Pirzada A. Timing matters: the interplay between early mealtime, circadian rhythms, gene expression, circadian hormones, and metabolism—a narrative review. *Clocks Sleep*. 2023;5:507-35.
 70. Phoi YY, Rogers M, Bonham MP, Dorrian J, Coates AM. A scoping review of chronotype and temporal patterns of eating of adults: tools used, findings, and future directions. *Nutr Res Rev*. 2022;35:112-35.
 71. Paoli A, Tinsley G, Bianco A, Moro T. The influence of meal frequency and timing on health in humans: the role of fasting. *Nutrients*. 2019;11:719.
 72. Cleator J, Abbott J, Judd P, Sutton C, Wilding JPH. Night eating syndrome: implications for severe obesity. *Nutr Diabetes*. 2012;2:e44.
 73. Westterp-Plantenga MS. Sleep, circadian rhythm and body weight: parallel developments. *Proc Nutr Soc*. 2016;75:431-9.
 74. Zéinski EL, Deibel SH, McDonald RJ. The trouble with circadian clock dysfunction: multiple deleterious effects on the brain and body. *Neurosci Biobehav Rev*. 2014;40:80-101.
 75. McClung CA. How might circadian rhythms control mood? Let me count the ways. *Biol Psychiatry*. 2013;74:242-9.
 76. Karatsoreos IN, Bhagat S, Bloss EB, Morrison JH, McEwen BS. Disruption of circadian clocks has ramifications for metabolism, brain, and behavior. *Proc Natl Acad Sci U S A*. 2011;108:1657-62.
 77. Drăgoi CM, Nicolae AC, Ungurianu A, Margină DM, Grădinaru D, Dumitrescu IB. Circadian rhythms, chrononutrition, physical training, and redox homeostasis—molecular mechanisms in human health. *Cells*. 2024;13:138.
 78. Papakonstantinou E, Oikonomou C, Nychas G, Dimitriadis GD. Effects of diet, lifestyle, chrononutrition and alternative dietary interventions on postprandial glycemia and insulin resistance. *Nutrients*. 2022;14:823.
 79. Challet E. The circadian regulation of food intake. *Nat Rev Endocrinol*. 2019;15:393-405.